Things I Don't Understand About Quantum Error Correction

A naive audit of the constructs — the confusions of a self-taught researcher, treated as data

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Poster Session: Wed 26 Aug, 16:30–18:00 · Rowan Quni-Gudzinas · Independent Researcher

Panel 1 — The Thesis: the constructs are cafeteria imports

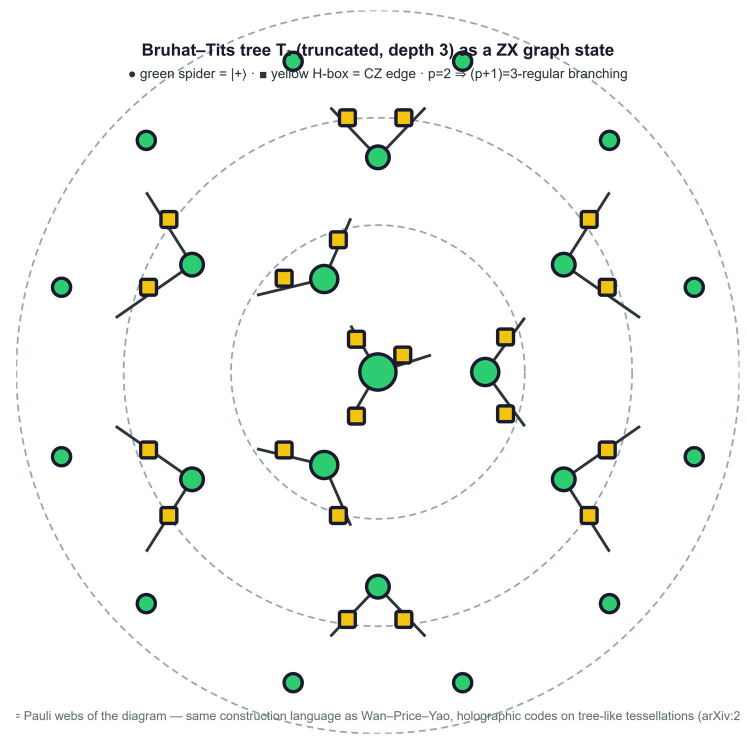
Every human construct for understanding quantum computing is **internally consistent** — and the imports **contradict each other**. Tools and systems are valid in their own silos; they were never checked for mutual compatibility.

Construct	Home silo	What it silently carries
Spiders	Categorical algebra (2D diagrammatic)	Drawing-plane topology; completeness proves facts about the *algebra*, nothing about spacetime
Pauli webs	Stabilizer formalism / QEC	2+1D code-over-time; in holographic use: entropy, wormholes, black holes
Gadgets	MBQC / compilation	Graph-state resources; rewrite soundness — not physical realization

Fault lines appear exactly where **3+1D particle physics and 1D thermodynamics** are mixed into a **2D map** without any compatibility check. The diagrams are not wrong — they are maps. The discipline of using a map is knowing where it ends.

Anchor: "ZX Diagrams at the Seam: Spiders, Pauli Webs, Gadgets, and the Cafeteria Problem of Cross-Disciplinary Imports" — 10.5281/zenodo.21992118 (2026-08-18).

Panel 2 — The Method: feigned naivete as probe



= Pauli webs of the diagram — same construction language as Wan–Price–Yao, holographic codes on tree-like tessellations (arXiv:2

I learned the room's language — the Bruhat–Tits tree T_2 (p=2, depth 3) as a ZX graph state: 22 spiders = 22 vertices, 21 H-boxes = 21 edges. **Learning the language is when the seams became visible.**

I ask the questions I'm not supposed to need to ask. The naive question is the **external structural probe**: it finds the locales and the seams. Where imports contradict, understanding breaks — **that break is data, not deficit.**

Method lineage: IAPS §6.2 audit-the-map; the Universal Ignorance Audit (15 questions, 10.5281/zenodo.21901984); "Knowing What We Do Not Know" (10.5281/zenodo.21901983).

Panel 3 — Things I don't understand about QEC

1 · Why the discreteness?
QEC protects a discrete abstraction (gates, qubits, circuits). Is the discreteness physical — or imported from the circuit model? If the abstraction is a choice, fault tolerance is the tax on the choice.

2 · What is a logical qubit, physically?
Hardware is bosonic oscillators with leakage levels; codes assume qubits. The map is 2D; the machine is continuous. Where does the map end?

3 · What does a threshold actually guarantee?
The proofs assume stochastic local noise. Coherent and correlated errors are the seams nobody prices. What is the guarantee outside the model?

4 · Where is the quantum in the loop?
Syndrome → *classical* decoder → feedback. QEC is quantum memory + classical control. Why is it called quantum error correction?

5 · Why isn't the decoder part of the code?
A classical algorithm decides a quantum code's performance. Coding theory and classical computation — two silos, never audited for compatibility.

Panel 3 (cont.) — The questions (2 of 2)

6 · Why 2D geometry?
Why a surface code and not a tree? Noise geometry is assumed continuous and Euclidean. What if error propagation is hierarchical — ultrametric (Bruhat–Tits)? The staircase nobody has looked for.

7 · Why count T gates, not joules?
The field prices the algebraic resource (T-count, magic states). Nobody prices the ancilla factory, verification retries, the decoder, the feedback. What does a correct answer cost in energy?

8 · Why is the cat the unit of protection?
Shor states are cat states — exponentially fragile to dephasing; verified ancillas go stale between verification and use. Why build protection from something that fragile?

Each question is a probe: the answer either closes the seam (the imports were compatible all along) or reveals it (the construct carried more than its proof).

References (verified 2026-08-18)
ZX Diagrams at the Seam — 10.5281/zenodo.21992118 (concept 10.5281/zenodo.21991895)
Universal Ignorance Audit — 10.5281/zenodo.21901984 · Knowing What We Do Not Know — 10.5281/zenodo.21901983
Ultrametric Code Spaces — 10.5281/zenodo.21824194 · JPCUB (joules-per-solution) — 10.5281/zenodo.21945415
Wan–Price–Yao, Holographic codes through ZX — arXiv:2601.04467 · van de Wetering — arXiv:2012.13966
Duncan–Kissinger–Perdrix–van de Wetering — arXiv:1902.03178 · Wan — arXiv:2410.17992

Rowan Quni-Gudzinas
Independent Researcher · Adelic Physics Programme
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